BUSINESS REQUIREMENTS DOCUMENT

Healthcare Patient & Hospital Analytics

SQL Analysis → Power BI Dashboard → GitHub Portfolio Project

Purpose: Define the business problem, analytical questions, KPIs, dashboard requirements and expected business outcomes. Requirements are derived directly from the supplied healthcare dataset and SQL analysis.

Item	Details
Source	healthcare.xlsx
Records	4,995
Period	2021–2024
Hospitals / Cities	15 / 5
Departments / Diagnoses	12 / 15
Tools	MySQL / SQL + Microsoft Power BI
Deliverable	Interactive Healthcare Analytics Dashboard

Core business question: How can hospital management use patient, admission, clinical, operational and financial data to understand demand, identify readmission patterns, monitor departmental performance and track revenue over time?

1. Business Problem

Healthcare organizations generate large volumes of patient and admission data, but raw transactional data does not directly provide management with a clear view of what is happening across time, departments, diagnoses and hospitals. Decision-makers need a consolidated analytical view to understand **when patients are admitted, who they are, why they are admitted, how long they stay, whether they return within 30 days, which departments experience higher demand, and how admission activity relates to revenue.**

The proposed solution will use SQL to extract and validate these insights and Power BI to transform them into an interactive management dashboard.

2. Business Objectives

- Track admission volume across 2021–2024 and identify yearly/monthly trends.
- Allow investigation of patients admitted during a selected month or year.
- Understand patient demographics through age and gender analysis.
- Identify diagnoses driving hospital admissions.
- Measure average patient length of stay.
- Measure 30-day readmission performance.
- Identify patients with multiple admissions.
- Compare admission workload across departments and years.
- Generate monthly reports containing admissions, unique patients and revenue.
- Understand how admission activity relates to hospital revenue.
- Enable users to answer these questions through Power BI without writing SQL.

3. Key Business Questions

Area	Business Question
Admission Trends	How many admissions occurred each year and month?
Monthly Monitoring	Which months have higher/lower admission volumes?
Patient Profile	What is the age and gender distribution?
Clinical Demand	Which diagnoses account for the highest admissions?
Department Performance	Which departments handle the highest admission volumes?
Length of Stay	What is the average patient length of stay?
Readmission	How many patients return within 30 days and what is the percentage?
Repeat Utilization	Which patients have more than one admission?
Revenue	How much revenue is generated by month/year and how does it move with admissions?

4. Analytical Scope

In scope: admission trends, demographics, diagnosis, length of stay, 30-day readmissions, repeat admissions, department analysis, monthly reporting, unique patients and revenue.

SQL operations not intended as dashboard requirements: UPDATE/DELETE statements and contact-information maintenance are database-operation exercises in the supplied SQL. They should not be performed on the portfolio source dataset.

5. SQL Analysis → Business Requirement Mapping

SQL Analysis	Dashboard / Business Requirement
Admissions in specific month	Year + Month filters and admission detail.
Min/max admission date	Define and validate the analysis period.
Admissions by year	Annual admissions KPI/trend.
Specific diagnosis patients	Diagnosis filtering and clinical demand analysis.
Female + selected conditions	Gender + diagnosis cross-analysis.
Patients sorted by age	Age distribution and detail analysis.
Average length of stay	Operational stay KPI.
Readmitted within 30 days	30-day readmission count and rate.
>1 admission	Repeat-admission analysis.
Department counts	Department workload comparison.
Department yearly pivot	Department-by-year matrix.
Monthly admission report	Monthly admissions + unique patients + revenue.

6. Recommended Dashboard Pages

- **01 Executive Overview:** Total Admissions, Unique Patients, Readmission %, Average Length of Stay, Revenue, admission trend and department/diagnosis highlights.
- **02 Patient & Clinical Analysis:** age, gender, diagnoses, selected-diagnosis patient view and length-of-stay analysis.
- **03 Admissions & Department Analysis:** monthly/yearly admissions, department-by-year comparison and hospital/department filters.
- **04 Readmission & Utilization:** 30-day readmissions, days between admissions, repeat-admission patients and diagnosis/hospital patterns.
- **05 Monthly Revenue & Performance:** monthly admissions, unique patients and revenue with trend analysis.
- **06 Patient Detail / Drill-through:** filtered patient-level view for authorized/internal analysis.

7. KPI Requirements

KPI	Definition
Total Admissions	Count of admission records in selected context.
Unique Patients	COUNT DISTINCT Patient_Name in selected context.
30-Day Readmissions	Readmission_Date within 30 days of Admission_Date.
30-Day Readmission %	Readmitted patients ÷ total patients × 100.
Average Length of Stay	AVERAGE(Avg_Length_of_Stay).
Monthly Revenue	SUM(Revenue) by Year + Month.
Annual Admissions	Admissions grouped by admission year.
Department Admissions	Admissions grouped by Department.
Repeat Admissions	Patients having more than one admission record.
Days Between	DATEDIFF(Readmission_Date, Admission_Date).

8. Required Dashboard Filters

Filter	Purpose
Year	Compare 2021–2024.
Month	Investigate monthly activity.
Admission Date	Flexible date range.
Hospital	Compare hospital performance.
City	Geographic comparison.
Department	Specialty workload.
Diagnosis	Clinical demand.
Gender	Demographic analysis.
Age / Age Group	Patient segmentation.

9. Data Model Requirements

- Create a dedicated Date dimension for Year, Quarter, Month and Month Number.
- Use the healthcare table as the primary source/fact table.
- Create explicit measures for admissions, unique patients, readmissions, readmission %, average stay and revenue.
- Sort month names chronologically using Month Number.
- Keep patient details on a dedicated drill-through page.
- Validate Power BI results against SQL results before publishing.

10. SQL Validation Requirements

Validation	Expected Check
Annual admissions	Power BI yearly totals reconcile with SQL GROUP BY year.
Monthly admissions	Power BI monthly trend reconciles with SQL monthly report.
Unique patients	Power BI distinct count reconciles with COUNT(DISTINCT).
30-day readmission	Power BI follows DATEDIFF <= 30 rule.
Department analysis	Department totals/yearly matrix reconcile with SQL.
Revenue	Monthly/yearly revenue reconciles with SUM(Revenue).

11. Data Quality & Logic Considerations

- Use proper NULL/date handling for Readmission_Date.
- Clearly document whether readmission percentage is patient-based or record-based.
- Use COUNT DISTINCT for unique-patient KPIs.
- Average Length of Stay must be averaged, not summed.
- Patient_Name may not be a reliable unique identifier in a real healthcare system; document this limitation.
- Do not expose patient names, contact information or addresses in public GitHub screenshots.
- Do not execute UPDATE/DELETE practice statements on the original portfolio dataset.

12. Expected Business Outcomes

After implementation, management should be able to answer the following without manually writing SQL:

- Are admissions increasing or decreasing over time?
- Which months experience the highest admission demand?
- Which departments receive the highest admission volumes?
- Which diagnoses are most common?
- What does the admitted patient population look like by age and gender?
- How long do patients typically stay?
- What percentage of patients are readmitted within 30 days?
- Which patients show repeat-admission behavior?
- How many unique patients are being served?
- How does monthly revenue move relative to admission volume?

13. Recommended Executive Layout

Zone	Content
Top KPI row	Admissions | Unique Patients | Readmission % | Avg Stay | Revenue
Trend section	Monthly admissions + annual comparison
Clinical section	Top diagnoses + department distribution
Utilization section	Repeat admissions + readmission analysis
Financial section	Monthly admissions vs revenue
Filter panel	Year | Month | Hospital | City | Department | Diagnosis | Gender | Age

14. Acceptance Criteria

ID	Acceptance Criterion
AC01	Dashboard covers the major business questions in this BRD.
AC02	Annual/monthly admissions reconcile with SQL.
AC03	Unique patient KPIs use distinct-patient logic.
AC04	30-day readmission follows the documented DATEDIFF rule.
AC05	Department-by-year analysis is available.
AC06	Monthly report contains admissions, unique patients and revenue.
AC07	Users can filter by Year and Month.
AC08	Users can drill from summary to relevant detail.
AC09	Measures respond correctly to filters.
AC10	Public portfolio materials exclude sensitive patient information.

15. GitHub Project Structure

File / Folder	Purpose
README.md	Business problem, solution, screenshots, KPIs, SQL and insights.
BRD/Healthcare_Analytics_BRD.pdf	Business Requirements Document.
SQL/Healthcare_Analysis.sql	SQL analysis and validation queries.
Dataset/healthcare.xlsx	Dataset, only if public redistribution is permitted.
PowerBI/Healthcare_Analytics.pbix	Final Power BI report.
Screenshots/	Dashboard screenshots for recruiter review.
Documentation/Data_Dictionary.md	Optional field documentation.

16. Portfolio Positioning

This should be presented as an **end-to-end Healthcare Analytics solution**, not simply a Power BI dashboard: **Business Problem → SQL Analysis → KPI Definition → Validation → Power BI Data Model → DAX Measures → Interactive Dashboard → Business Insights.**

17. Definition of Done

- SQL script is organized around business questions.
- BRD clearly documents the business problem and analytical scope.
- Power BI model is built from validated source data.
- Core KPIs are implemented as explicit measures.
- Annual/monthly admissions, clinical, demographic and department analysis are available.
- 30-day readmission and repeat-admission analysis are available.
- Monthly revenue is analyzed alongside admission activity.
- Dashboard has intuitive filters, navigation and drill-through.
- GitHub contains BRD + SQL + PBIX/screenshots + README.
- Sensitive patient information is not publicly exposed.

Prepared from the supplied healthcare.xlsx dataset and SQL requirements • August 2026